University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

Operating Systems (OS) E1
MID TERM

Paper Duration: 90 minutes **Total Marks: 40**

INSTRUCTIONS:

- Write your Name, Registration No and Section in the provided space.
- 70% deduction in case of **compilation error**.
- 50% deduction in case of run time error (**segmentation fault**).

Question 1: Submit it on Portal along with screenshots and code. (20 Marks)

Write a C++ program that uses the **`fork()` system call** to create a **child process**. Both the parent and the child should perform different tasks.

Requirements:

- The **parent process** accepts a number `n` from the user and prints the multiplication table of `n`.
- The **child process** calculates the factorial of the same number `n` and displays the result.
- Both processes should clearly display which one is executing (e.g., "Parent process output" or "Child process output").
- Use the `getpid()` and `getppid()` functions to display the process IDs of the parent and child.

Sample Output:

```
Enter a number: 5

Parent Process (PID: 1205)
Multiplication table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
```

```
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Child Process (PID: 1206, PPID: 1205)
Factorial of 5 = 120
```

Note: Attach the screenshots of your terminal. It carries **2 marks.**

Question 2: Submit it on Portal along with screenshots and code. **(20 Marks)**

Problem Statement:

Write a C++ program that uses **pipes** to enable communication between a parent and a child process created with `fork()`.

Requirements:

- The **parent process** takes a string input from the user.
- It sends this string to the **child process** through a **pipe**.
- The **child process** checks whether the string is a palindrome (reads the same backward).
    - If it is a palindrome, the child sends the message: `"The string is a palindrome"` back to the parent.
    - Otherwise, it sends the **reversed string** back to the parent.
- The **parent process** displays the message or reversed string received from the child.

Sample output:

```
Enter a string: madam
Child: The string is a palindrome.
Parent: Message received from child.

Enter a string: hello
Child: Reversed string is: olleh
Parent: Message received from child.
```

Note: Attach the screenshots of your terminal. It carries **2 marks.**